

06_Assignment_Yuliia_Chernysheva

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CSC 2267

```
[1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker
```

```
[2]: # Load Dataset and Initial Inspection
df_inequality_gdp = pd.read_csv("gini-vs-gdp-per-capita.csv")
df_inequality_gdp.head()
```

```
[2]:
```

	Entity	Code	Year	Gini index \	
0	Abkhazia	OWID_ABK	2015	NaN	
1	Afghanistan	AFG	2002	NaN	
2	Afghanistan	AFG	2003	NaN	
3	Afghanistan	AFG	2004	NaN	
4	Afghanistan	AFG	2005	NaN	

	GDP per capita, PPP (constant 2017 international \$)	Continent
0	NaN	Asia
1	1189.784668	NaN
2	1235.810063	NaN
3	1200.278013	NaN
4	1286.793659	NaN

```
[3]: # Rename Columns
names_dict = {'Gini index': 'Gini_ind',
              'GDP per capita, PPP (constant 2017 international $)':
               ↪ 'GDP_per_capita'}
df_selected = df_inequality_gdp.rename(columns=names_dict)
df_selected
```

```
[3]:
```

	Entity	Code	Year	Gini_ind	GDP_per_capita	Continent
0	Abkhazia	OWID_ABK	2015	NaN	NaN	Asia
1	Afghanistan	AFG	2002	NaN	1189.784668	NaN
2	Afghanistan	AFG	2003	NaN	1235.810063	NaN

3	Afghanistan	AFG	2004	NaN	1200.278013	NaN
4	Afghanistan	AFG	2005	NaN	1286.793659	NaN
...	...	...	...	...	...	...
8684	Zimbabwe	ZWE	2015	NaN	3198.982129	Africa
8685	Zimbabwe	ZWE	2016	NaN	3173.610829	NaN
8686	Zimbabwe	ZWE	2018	NaN	3341.665418	NaN
8687	Zimbabwe	ZWE	2020	NaN	2744.690758	NaN
8688	Åland Islands	ALA	2015	NaN	NaN	Europe

[8689 rows x 6 columns]

```
[4]: # Clean Missing Values (Drop rows where both "Gini_ind", "GDP_per_capita" are
      ↪NaN)
df_selected = df_selected.dropna(subset=["Gini_ind", "GDP_per_capita"],
      ↪how="all")
df_selected
```

```
[4]:
```

	Entity	Code	Year	Gini_ind	GDP_per_capita	Continent
1	Afghanistan	AFG	2002	NaN	1189.784668	NaN
2	Afghanistan	AFG	2003	NaN	1235.810063	NaN
3	Afghanistan	AFG	2004	NaN	1200.278013	NaN
4	Afghanistan	AFG	2005	NaN	1286.793659	NaN
5	Afghanistan	AFG	2006	NaN	1315.789117	NaN
...	...	...	...	...	...	...
8683	Zimbabwe	ZWE	2014	NaN	3195.767970	NaN
8684	Zimbabwe	ZWE	2015	NaN	3198.982129	Africa
8685	Zimbabwe	ZWE	2016	NaN	3173.610829	NaN
8686	Zimbabwe	ZWE	2018	NaN	3341.665418	NaN
8687	Zimbabwe	ZWE	2020	NaN	2744.690758	NaN

[8600 rows x 6 columns]

```
[5]: # Check Dataset Information & Memory Usage
df_selected.info(memory_usage='deep')
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 8600 entries, 1 to 8687
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Entity          8600 non-null   object
1   Code            7033 non-null   object
2   Year            8600 non-null   int64
3   Gini_ind        4384 non-null   float64
4   GDP_per_capita  7109 non-null   float64
5   Continent       196 non-null    object
dtypes: float64(2), int64(1), object(3)
```

memory usage: 1.5 MB

```
[6]: # Extract Unique Entities List
     entities_list = df_selected['Entity'].unique().tolist()
     print(entities_list)
```

```
['Afghanistan', 'Africa Eastern and Southern', 'Africa Western and Central',
 'Albania', 'Algeria', 'Angola', 'Antigua and Barbuda', 'Arab World',
 'Argentina', 'Argentina (Urban)', 'Armenia', 'Aruba', 'Australia', 'Austria',
 'Azerbaijan', 'Bahamas', 'Bahrain', 'Bangladesh', 'Barbados', 'Belarus',
 'Belgium', 'Belize', 'Benin', 'Bermuda', 'Bhutan', 'Bolivia', 'Bosnia and
 Herzegovina', 'Botswana', 'Brazil', 'Brunei', 'Bulgaria', 'Burkina Faso',
 'Burundi', 'Cambodia', 'Cameroon', 'Canada', 'Cape Verde', 'Caribbean Small
 States', 'Cayman Islands', 'Central African Republic', 'Central Europe and the
 Baltics', 'Chad', 'Chile', 'China', 'China (Rural)', 'China (Urban)',
 'Colombia', 'Comoros', 'Congo', 'Costa Rica', 'Cote d'Ivoire', 'Croatia',
 'Curacao', 'Cyprus', 'Czechia', 'Democratic Republic of Congo', 'Denmark',
 'Djibouti', 'Dominica', 'Dominican Republic', 'Early-demographic dividend',
 'East Asia & Pacific', 'East Asia & Pacific (IDA & IBRD)', 'East Asia & Pacific
 (excluding high income)', 'Ecuador', 'Egypt', 'El Salvador', 'Equatorial
 Guinea', 'Estonia', 'Eswatini', 'Ethiopia', 'Euro area', 'Europe & Central
 Asia', 'Europe & Central Asia (IDA & IBRD)', 'Europe & Central Asia (excluding
 high income)', 'European Union', 'Fiji', 'Finland', 'France', 'Gabon', 'Gambia',
 'Georgia', 'Germany', 'Ghana', 'Greece', 'Grenada', 'Guatemala', 'Guinea',
 'Guinea-Bissau', 'Guyana', 'Haiti', 'Heavily indebted poor countries (HIPC)',
 'High income', 'Honduras', 'Hong Kong', 'Hungary', 'IBRD only', 'IDA & IBRD
 total', 'IDA blend', 'IDA only', 'IDA total', 'Iceland', 'India', 'India
 (Rural)', 'India (Urban)', 'Indonesia', 'Indonesia (Rural)', 'Indonesia
 (Urban)', 'Iran', 'Iraq', 'Ireland', 'Israel', 'Italy', 'Jamaica', 'Japan',
 'Jordan', 'Kazakhstan', 'Kenya', 'Kiribati', 'Kosovo', 'Kuwait', 'Kyrgyzstan',
 'Laos', 'Late-demographic dividend', 'Latin America & Caribbean', 'Latin America
 & Caribbean (IDA & IBRD)', 'Latin America & Caribbean (excluding high income)',
 'Latvia', 'Least developed countries: UN classification', 'Lebanon', 'Lesotho',
 'Liberia', 'Libya', 'Lithuania', 'Low & middle income', 'Lower middle income',
 'Luxembourg', 'Macao', 'Madagascar', 'Malawi', 'Malaysia', 'Maldives', 'Mali',
 'Malta', 'Marshall Islands', 'Mauritania', 'Mauritius', 'Mexico', 'Micronesia
 (country)', 'Middle East & North Africa', 'Middle East & North Africa (IDA &
 IBRD)', 'Middle East & North Africa (excluding high income)', 'Middle income',
 'Moldova', 'Mongolia', 'Montenegro', 'Morocco', 'Mozambique', 'Myanmar',
 'Namibia', 'Nauru', 'Nepal', 'Netherlands', 'New Zealand', 'Nicaragua', 'Niger',
 'Nigeria', 'North America', 'North Macedonia', 'Norway', 'OECD members', 'Oman',
 'Other small states', 'Pacific island small states', 'Pakistan', 'Palau',
 'Palestine', 'Panama', 'Papua New Guinea', 'Paraguay', 'Peru', 'Philippines',
 'Poland', 'Portugal', 'Post-demographic dividend', 'Pre-demographic dividend',
 'Puerto Rico', 'Qatar', 'Romania', 'Russia', 'Rwanda', 'Saint Kitts and Nevis',
 'Saint Lucia', 'Saint Vincent and the Grenadines', 'Samoa', 'San Marino', 'Sao
 Tome and Principe', 'Saudi Arabia', 'Senegal', 'Serbia', 'Seychelles', 'Sierra
 Leone', 'Singapore', 'Sint Maarten (Dutch part)', 'Slovakia', 'Slovenia', 'Small
```

```
states', 'Solomon Islands', 'Somalia', 'South Africa', 'South Asia', 'South Asia (IDA & IBRD)', 'South Korea', 'South Sudan', 'Spain', 'Sri Lanka', 'Sub-Saharan Africa', 'Sub-Saharan Africa (IDA & IBRD)', 'Sub-Saharan Africa (excluding high income)', 'Sudan', 'Suriname', 'Suriname (Urban)', 'Sweden', 'Switzerland', 'Syria', 'Taiwan', 'Tajikistan', 'Tanzania', 'Thailand', 'Timor', 'Togo', 'Tonga', 'Trinidad and Tobago', 'Tunisia', 'Turkey', 'Turkmenistan', 'Turks and Caicos Islands', 'Tuvalu', 'Uganda', 'Ukraine', 'United Arab Emirates', 'United Kingdom', 'United States', 'Upper middle income', 'Uruguay', 'Uzbekistan', 'Vanuatu', 'Venezuela', 'Vietnam', 'World', 'Yemen', 'Zambia', 'Zimbabwe']
```

```
[7]: # Filter Out Non-Country Entities (Regions, Income Groups, etc.)
not_countries = [
    "Africa Eastern and Southern", "Africa Western and Central", "Arab World",
    "Argentina (Urban)", "Aruba", "Bermuda", "Caribbean Small States",
    "Cayman Islands", "Central Central Europe and the Baltics", "China (Rural)",
    "China (Urban)", "Curacao", "Early-demographic dividend", "East Asia &
    ↪Pacific",
    "East Asia & Pacific (IDA & IBRD)", "East Asia & Pacific (excluding high
    ↪income)",
    "Euro area", "Europe & Central Asia", "Europe & Central Asia (IDA & IBRD)",
    "Europe & Central Asia (excluding high income)", "European Union",
    "Heavily indebted poor countries (HIPC)", "High income", "Hong Kong",
    "IBRD only", "IDA & IBRD total", "IDA blend", "IDA only", "IDA total",
    "India (Rural)", "India (Urban)", "Indonesia (Rural)", "Indonesia (Urban)",
    "Latin America & Caribbean", "Latin America & Caribbean (IDA & IBRD)",
    "Latin America & Caribbean (excluding high income)", "Late-demographic
    ↪dividend",
    "Least developed countries: UN classification", "Low & middle income",
    "Lower middle income", "Macao", "Middle East & North Africa",
    "Middle East & North Africa (IDA & IBRD)", "Middle East & North Africa
    ↪(excluding high income)",
    "Middle income", "North America", "OECD members", "Other small states",
    "Pacific island small states", "Palestine", "Post-demographic dividend",
    "Pre-demographic dividend", "Puerto Rico", "Sint Maarten (Dutch part)",
    "Small states", "South Asia", "South Asia (IDA & IBRD)", "Sub-Saharan
    ↪Africa",
    "Sub-Saharan Africa (IDA & IBRD)", "Sub-Saharan Africa (excluding high
    ↪income)",
    "Suriname (Urban)", "Turks and Caicos Islands", "Upper middle income",
    ↪"World"
]

df_selected_countries = df_selected[~df_selected.Entity.isin(not_countries)]
df_selected_countries.head()
```

```
[7]:      Entity Code  Year  Gini_ind  GDP_per_capita  Continent
1  Afghanistan  AFG  2002      NaN    1189.784668      NaN
```

2	Afghanistan	AFG	2003	NaN	1235.810063	NaN
3	Afghanistan	AFG	2004	NaN	1200.278013	NaN
4	Afghanistan	AFG	2005	NaN	1286.793659	NaN
5	Afghanistan	AFG	2006	NaN	1315.789117	NaN

```
[8]: # Drop Unnecessary Columns: Year, Continent, Code if they exist
drop_columns = ['Code', 'Continent']

df_selected_countries = df_selected_countries.drop(
    columns=drop_columns,
    errors='ignore'
)
df_selected_countries.head()
```

```
[8]:      Entity  Year  Gini_ind  GDP_per_capita
1  Afghanistan  2002      NaN    1189.784668
2  Afghanistan  2003      NaN    1235.810063
3  Afghanistan  2004      NaN    1200.278013
4  Afghanistan  2005      NaN    1286.793659
5  Afghanistan  2006      NaN    1315.789117
```

```
[9]: # Sort the data by country and year to ensure the correct time-series order
df_sorted = df_selected_countries.sort_values(['Entity', 'Year'])

# Create a copy for filling in the blanks
df_filled = df_sorted.copy()
df_filled.head()
```

```
[9]:      Entity  Year  Gini_ind  GDP_per_capita
1  Afghanistan  2002      NaN    1189.784668
2  Afghanistan  2003      NaN    1235.810063
3  Afghanistan  2004      NaN    1200.278013
4  Afghanistan  2005      NaN    1286.793659
5  Afghanistan  2006      NaN    1315.789117
```

```
[10]: # Fill missing values with the last known value (ffill)
# and the previous one (bfill) for each country separately
df_filled['Gini_ind'] = (
    df_filled.groupby('Entity')['Gini_ind'].ffill().bfill()
)
df_filled['GDP_per_capita'] = (
    df_filled.groupby('Entity')['GDP_per_capita'].ffill().bfill()
)
df_filled.head()
```

```
[10]:      Entity  Year  Gini_ind  GDP_per_capita
1  Afghanistan  2002  27.01034    1189.784668
```

2	Afghanistan	2003	27.01034	1235.810063
3	Afghanistan	2004	27.01034	1200.278013
4	Afghanistan	2005	27.01034	1286.793659
5	Afghanistan	2006	27.01034	1315.789117

```
[11]: # Get the count of missing values
missing_count = df_filled.isnull().sum()
print(missing_count)
```

```
Entity      0
Year        0
Gini_ind    0
GDP_per_capita  0
dtype: int64
```

```
[12]: selected_year = 2019

# Take a snapshot based strictly on the target year (2019).
df_2019 = df_filled[df_filled['Year'] == selected_year].copy()
df_2019
```

```
[12]:
```

	Entity	Year	Gini_ind	GDP_per_capita
18	Afghanistan	2019	27.01034	2065.036235
108	Albania	2019	33.17109	13671.488422
139	Algeria	2019	27.61573	11510.557088
183	Angola	2019	51.27211	6670.331458
235	Antigua and Barbuda	2019	44.41862	21548.725008
...	...	...	...	...
8480	Vanuatu	2019	37.62623	3117.678832
8525	Vietnam	2019	35.71555	8041.178384
8603	Yemen	2019	36.70709	2190.181724
8626	Zambia	2019	57.13606	3470.448024
8680	Zimbabwe	2019	50.25645	3027.656038

```
[189 rows x 4 columns]
```

```
[13]: # Remove countries that still have NaNs (those that had no data at all)
df_sel_2019 = df_2019.dropna(subset=['Gini_ind', 'GDP_per_capita'], how='any')

# Check the result
print(f"Number of countries with complete data for 2019: {len(df_sel_2019)}\n")
df_sel_2019.head()
```

```
Number of countries with complete data for 2019: 189
```

```
[13]:
```

	Entity	Year	Gini_ind	GDP_per_capita
18	Afghanistan	2019	27.01034	2065.036235
108	Albania	2019	33.17109	13671.488422
139	Algeria	2019	27.61573	11510.557088
183	Angola	2019	51.27211	6670.331458
235	Antigua and Barbuda	2019	44.41862	21548.725008

```
[14]: # Statistical Summary of Cleaned Data
df_sel_2019.describe().T
```

```
[14]:
```

	count	mean	std	min	25%	\
Year	189.0	2019.000000	0.000000	2019.000000	2019.000000	
Gini_ind	189.0	38.249112	8.578614	20.186580	32.380690	
GDP_per_capita	189.0	20255.673200	20592.241228	751.664153	4388.800909	

	50%	75%	max
Year	2019.000000	2019.000000	2019.000000
Gini_ind	37.054930	43.71005	63.026070
GDP_per_capita	13070.125645	28753.51627	113940.237442

```
[15]: display(
    df_sel_2019[df_sel_2019.Entity.isin(['Gabon', 'United States'])].style.
    ↪hide(axis='index')
)
```

<pandas.io.formats.style.Styler at 0x7a97d2b387f0>

```
[16]: # Visualize Income Inequality vs GDP per Capita for 2019
plt.figure(figsize=(12,7))

sns.scatterplot(
    data=df_sel_2019,
    x='GDP_per_capita',
    y='Gini_ind',
    hue='Entity',
    size='Gini_ind',
    legend=False,
    alpha=0.8)

# Enable a logarithmic scale for the X-axis, as in the original
plt.xscale('log')

# Headings
plt.suptitle("Income inequality vs GDP per capita, 2019",
             fontsize=16, fontweight='bold');
plt.title("A higher Gini index indicates higher inequality",
          fontsize=12, style='italic')
```

```

plt.xlabel("GDP per capita", fontsize=14)
plt.ylabel("Gini index", fontsize=14)

# Format the X-axis with dollar signs and thousands separators
ax = plt.gca()
ax.xaxis.set_major_formatter(ticker.StrMethodFormatter('${x:,.0f}'))

# Specify specific values on the X-axis so the tick marks look neat
ticks = [1000, 2000, 5000, 10000, 20000, 50000, 100000]
ax.set_xticks(ticks)
ax.set_xticklabels([f'${t:,.0f}' for t in ticks])

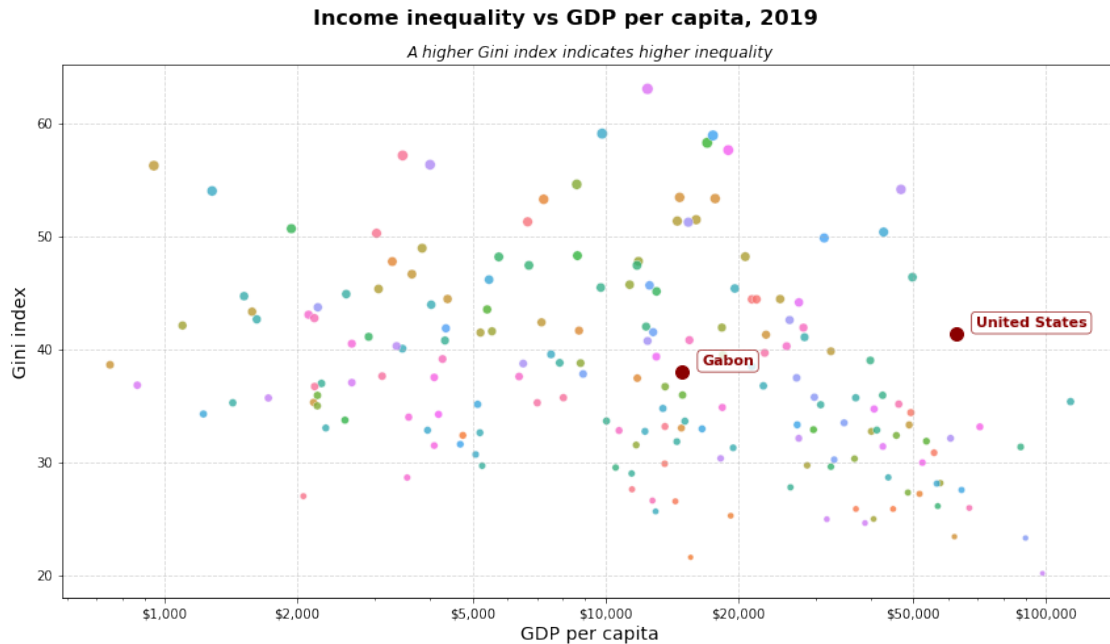
# Highlight Gabon and the United States (as required in the PDF examples)
highlight_countries = ['Gabon', 'United States']
highlight_df = df_sel_2019[df_sel_2019['Entity'].isin(highlight_countries)]

for _, row in highlight_df.iterrows():
    # Draw a standout point for these countries
    plt.scatter(row['GDP_per_capita'], row['Gini_ind'], color='darkred', s=100,
        ↪zorder=5)

    # Adding a text label next to the point
    plt.annotate(
        row['Entity'],
        xy=(row['GDP_per_capita'], row['Gini_ind']),
        xytext=(15, 5), textcoords='offset points',
        fontweight='bold', fontsize=11, color='darkred',
        bbox=dict(boxstyle="round,pad=0.3", fc="white", ec="darkred", lw=0.8,
        ↪alpha=0.8)
    )

plt.grid(True, which="major", ls="--", alpha=0.5)
plt.tight_layout()
plt.show()

```



0.0.1 Interpretation of Results

The scatter plot shows a weak negative relationship between GDP per capita and income inequality. In general, countries with higher GDP per capita tend to have somewhat lower Gini index values, although there is considerable variation among countries. For example, the United States has a high GDP per capita but also a relatively high level of income inequality. Gabon also illustrates that a higher GDP per capita does not necessarily mean that income is distributed equally. Overall, the graph suggests that economic development may be associated with lower inequality, but GDP per capita alone does not determine how equally income is distributed. Government policies and the distribution of wealth also play an important role.

[]: